System Verilog Projects

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Project-1: Smart Traffic Controller Verification using SystemVerilog

Project Description:

Developed a four-way Smart Traffic Controller using SystemVerilog and a Finite State Machine (FSM). The controller manages traffic signals for four roads and handles pedestrian crossing requests and emergency vehicle priority. Developed a class-based testbench to generate input transactions, drive signals, monitor outputs, and check the traffic controller's operation.

Key Features:

- Four-way traffic signal control.
- Red, Yellow, and Green signal management.
- Emergency vehicle priority.
- Pedestrian crossing request handling.
- Clock-based FSM state transitions.
- Generator and Driver communication using a mailbox.
- Monitor to observe traffic signal outputs.
- Scoreboard to check that multiple roads are not GREEN simultaneously.
- VCD waveform generation for simulation analysis.

Tools Used: SystemVerilog, EDA Playground.

Verification: Verified normal traffic operation, pedestrian requests, and emergency requests using a class-based testbench. Used a monitor and scoreboard to check traffic signal outputs and ensure that multiple roads are not GREEN at the same time.

EDA Link: https://edaplayground.com/x/VMkP

Project-2: 8b/10b Encoder Design and Verification using SystemVerilog

Project Description:

Developed an 8b/10b Encoder using SystemVerilog to convert 8-bit input data into a 10-bit encoded output. The design uses 5b/6b and 3b/4b encoding techniques and supports running disparity management. Developed a class-based testbench to generate input transactions, drive the DUT, monitor outputs, and compare the actual results with expected values using a reference model.

Key Features:

- Converts 8-bit data into 10-bit encoded output.
- Implements 5b/6b and 3b/4b encoding.
- Supports positive and negative running disparity.
- Updates running disparity after each encoding stage.
- Generates 512 test transactions.
- Uses Generator, Driver, Monitor, and Scoreboard components.
- Uses a reference model to calculate expected outputs.
- Checks encoded data, running disparity, and valid signals.
- Provides a PASS and FAIL summary.

Tools Used: SystemVerilog, EDA Playground.

Verification: Verified the encoder using 512 transactions covering all 256 input data values with both running disparity inputs (0 and 1). Used a reference model and scoreboard to compare expected and actual encoded outputs and check running disparity and valid signals.

EDA Link: https://edaplayground.com/x/PUbt

Project-3: Washing Machine Controller Design and Verification using SystemVerilog

Project Description:

Developed a Washing Machine Controller using SystemVerilog and a Finite State Machine (FSM). The controller manages the complete washing process through different stages, including filling, washing, rinsing, and spinning. It uses water-level detection, timer-based state transitions, and safety conditions such as door-open and stop signals. Developed a testbench to apply inputs, monitor FSM transitions, and observe the output signals.

Key Features:

- FSM-based washing machine controller.
- Six states: IDLE, FILL, WASH, RINSE, SPIN, and DONE.
- Start signal to begin the washing cycle.
- Water-level detection before washing.
- Timer-based transitions between washing stages.
- Door-open and stop conditions to interrupt the cycle.
- Controls water valve, wash motor, rinse valve, and spin motor.
- Buzzer and done signals to indicate cycle completion.
- VCD waveform generation for simulation analysis.

Tools Used: SystemVerilog, EDA Playground.

Verification: Verified the washing machine operation using a SystemVerilog testbench. Tested reset, start, and water-level inputs, monitored FSM state transitions, and observed the control outputs for each stage of the washing cycle.

EDA Link: https://edaplayground.com/x/Vst3

Project-4: Half Adder Design and Verification using SystemVerilog

Project Description:

Designed and verified a Half Adder using SystemVerilog. The Half Adder performs the addition of two 1-bit binary inputs and generates Sum and Carry outputs. Developed a simple class-based testbench using a transaction class to apply all possible input combinations and check the outputs.

Key Features:

- Half Adder design using XOR and AND logic gates.
- Generates Sum and Carry outputs.
- Uses a transaction class to store input and output values.
- Tests all four possible input combinations.
- Uses a scoreboard to compare expected and actual outputs.
- Displays PASS or FAIL for each test case.
- Generates a VCD waveform file for simulation analysis.

Tools Used: SystemVerilog, EDA Playground.

Verification: Verified all four input combinations using a SystemVerilog testbench. Checked the Sum and Carry outputs against their expected values using scoreboard-based checking.

EDA Link: https://edaplayground.com/x/XY3b